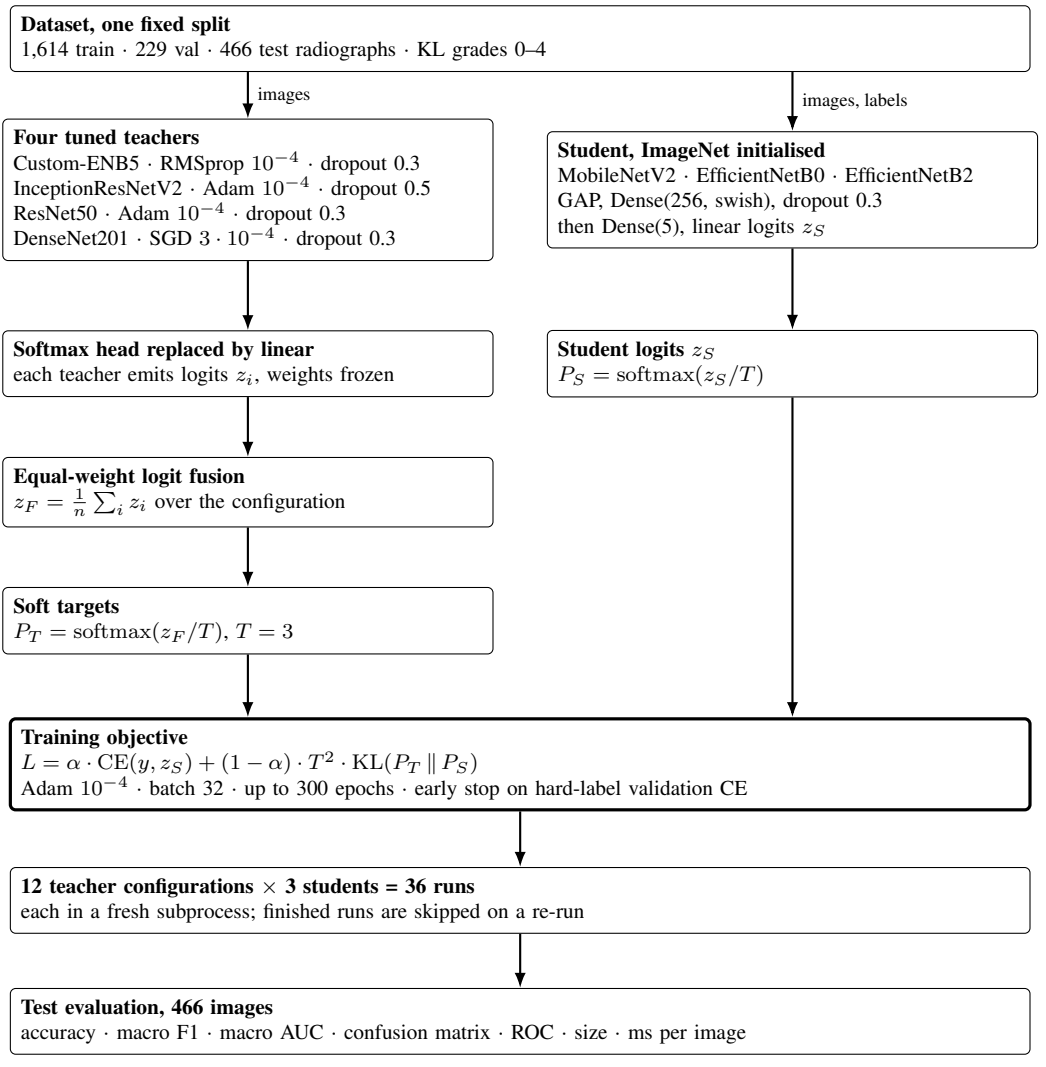
Dataset, one fixed split
1,614 train · 229 val · 466 test radiographs · KL grades 0–4
images
images, labels
Four tuned teachers
Custom-ENB5 · RMSprop 10^{-4} · dropout 0.3
InceptionResNetV2 · Adam 10^{-4} · dropout 0.5
ResNet50 · Adam 10^{-4} · dropout 0.3
DenseNet201 · SGD $3 \cdot 10^{-4}$ · dropout 0.3
Student, ImageNet initialised
MobileNetV2 · EfficientNetB0 · EfficientNetB2
GAP, Dense(256, swish), dropout 0.3
then Dense(5), linear logits z_S
Softmax head replaced by linear
each teacher emits logits z_i, weights frozen
Student logits z_S
$P_S = \mathrm{softmax}(z_S/T)$
Equal-weight logit fusion
$z_F = \frac{1}{n}\sum_i z_i$ over the configuration
Soft targets
$P_T = \mathrm{softmax}(z_F/T)$, $T = 3$
Training objective
$L = \alpha \cdot \mathrm{CE}(y, z_S) + (1-\alpha) \cdot T^2 \cdot \mathrm{KL}(P_T \,\|\, P_S)$
Adam 10^{-4} · batch 32 · up to 300 epochs · early stop on hard-label validation CE
12 teacher configurations × 3 students = 36 runs
each in a fresh subprocess; finished runs are skipped on a re-run
Test evaluation, 466 images
accuracy · macro F1 · macro AUC · confusion matrix · ROC · size · ms per image